

To build the project, we require the components mentioned under the Bill of

Materials table. We will also need some other parts for making this robot. Check

the link <https://www.electronicsforu.com/electronics-projects/design-iot-face-recognition-ai-robot> for that.

This project has two distinct features, for which we need to create a basic setup for both face recognition and deep dreams.

For face recognition requirements/setup, we need to set up the following libraries in Raspberry Pi:

- Keras
- OpenCV
- eSpeak
- Dlib
- Face recognition

To install the above libraries, open the terminal window in Raspberry Pi and run:

```
sudo apt-get update
sudo apt-get upgrade
sudo apt-get install build-essential
cmake pkg-config
sudo apt-get install libjpeg-dev
libtiff5-dev libjasper-dev libpng12-dev
sudo apt-get install libavcodec-dev
```

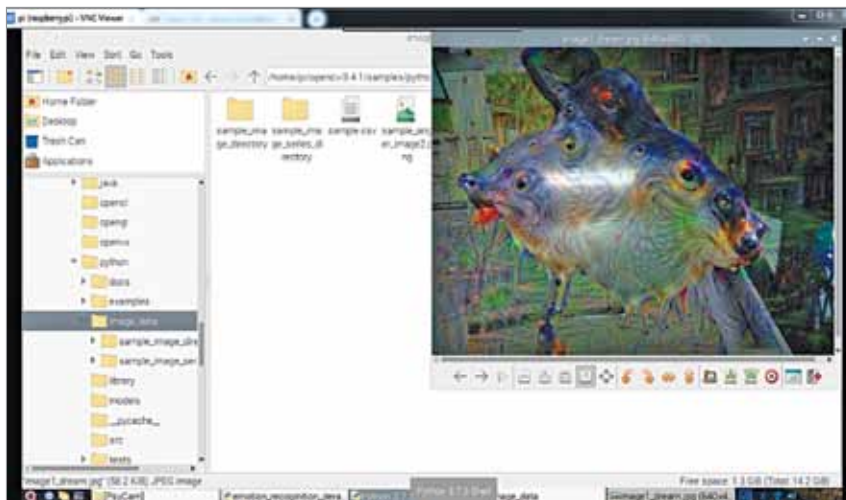


Figure 3: Creating a dream



Figure 4: Installing dlib

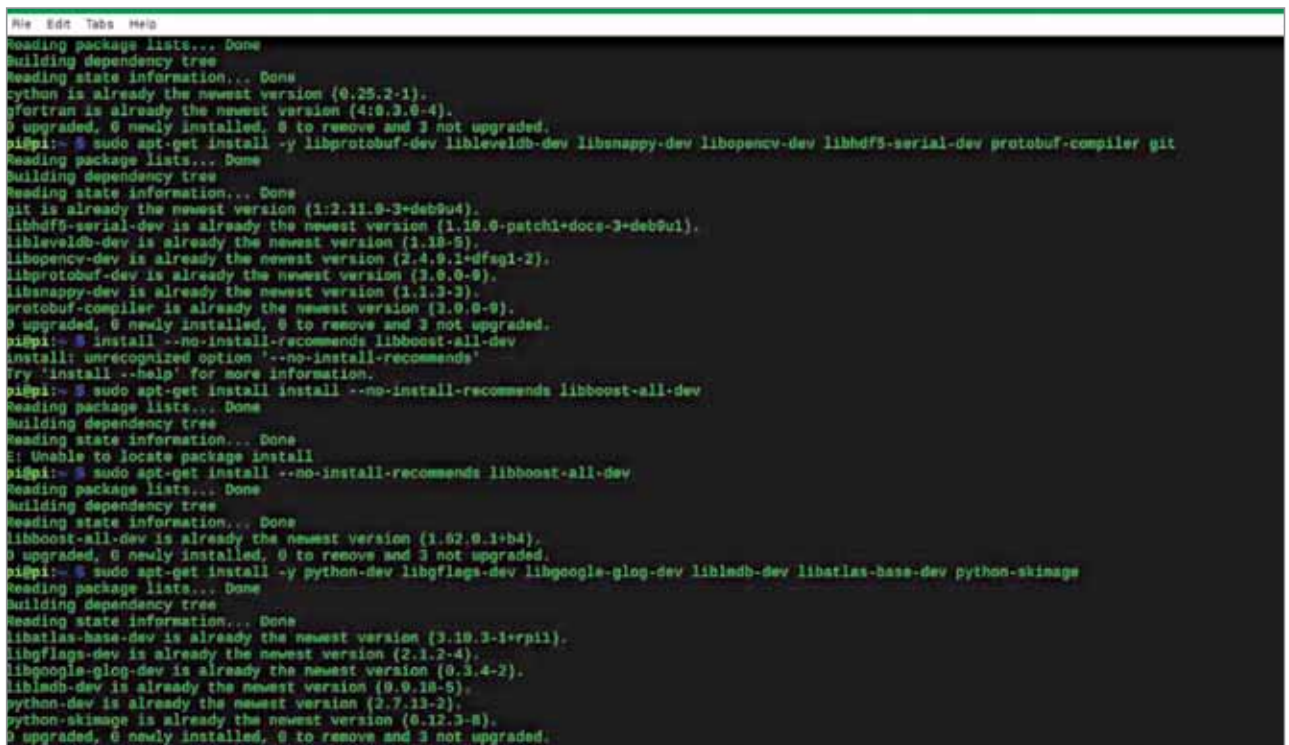


Figure 5: Installing Python modules